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The Expanding Energy Demand of Artificial Intelligence Data Centres: Societal Costs, Energy Poverty, and the European Union Policy Response

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Abstract: The rapid diffusion of artificial intelligence (AI) is adding a fast-growing workload to the data centres that host it, with consequences for electricity systems, household energy costs, and the environment. This article examines the sustainability implications of AI-driven data-centre expansion in Europe along three connected dimensions—environmental sustainability, energy and social equity, and the sustainability of the digital transition—and asks who gains and who pays. Doctrinal analysis of European Union (EU) law and policy was combined with a comparative, descriptive assessment of published data. The study synthesised evidence from international agencies, EU institutions, national statistical offices, regulators, and peer-reviewed research, and traced household electricity costs and data-centre development from 2023 onward in three purposively selected cases: Ireland, the United Kingdom, and Spain. Data-centre electricity demand, of which AI workloads are an increasingly important component, is growing far faster than overall demand and is highly concentrated geographically. In exposed regions this concentration is associated with grid-capacity constraints and may contribute to upward pressure on network and system costs. Household prices in the period were, however, shaped predominantly by gas prices, taxation, subsidies, renewable output, and network charges, and the study does not estimate causal effects. The expansion also generates environmental externalities, notably carbon emissions and a growing stream of electronic waste. Because energy costs weigh most heavily on low-income households, these pressures carry a distributional risk: a potential "double burden" in which households already in energy poverty benefit least from AI while bearing part of its infrastructure costs. The article maps the EU response by legal status—measures in force, formally proposed, and under discussion—and argues that assessments of AI's social value should internalise its energy, climate, resource, and distributional costs. It closes with actor-specific recommendations on clean-supply additionality, cost-reflective network-cost allocation, safeguards for vulnerable consumers, hardware circularity, and sustainability metrics.

Keywords: Artificial intelligence; Data centres; Electricity prices; Energy poverty; Energy justice; Electronic waste; Sustainable digital transition; European Union energy law

1. Introduction

Artificial intelligence (AI) promises to transform daily life, from healthcare to public administration. This transformation carries an energy cost. Data centres—the physical infrastructure of cloud computing and, increasingly, of AI—consumed about 415 terawatt-hours (TWh) in 2024, roughly 1.5% of global electricity. The International Energy Agency (IEA) projects that their consumption will more than double by 2030, to around 945 TWh or just under 3% of global electricity (IEA, 2025). Accelerated servers dedicated to AI workloads are the fastest-growing component of this demand (de Vries, 2023; IEA, 2025). The resulting questions concern not only the climate footprint of computing (Crawford, 2024; Kaack et al., 2022; Masanet et al., 2020) but also the resilience of electricity systems and the fairness with which their costs are shared.

Electricity is a networked good whose costs are socialised through tariffs and levies. When large new loads connect to a grid faster than generation and network capacity can be expanded, the resulting costs—new capacity, grid reinforcement, and balancing services—are recovered from all users, including households (Martin & Peskoe, 2025). This raises a distributional question that runs through the article: who gains and who pays? The commercial benefits of AI accrue first to technology firms and to the businesses and consumers able to use their services. The costs of hosting the infrastructure are borne locally and, through tariffs, collectively. For households already at risk of energy poverty, even modest additional pressure on bills can be significant (Bouzarovski et al., 2021; Guan et al., 2023). The possibility that such households gain least from AI while bearing part of its infrastructure costs is referred to here as a potential "double burden". Throughout the article, this possibility is treated as a distributional risk to be examined rather than as an effect that the study demonstrates.

Seen in this light, the expansion of AI-related computing is a sustainability question in three connected senses. The first is environmental sustainability: the carbon emissions associated with the electricity consumed and embodied in the hardware, and the growing stream of electronic waste (Gupta et al., 2022; Wang et al., 2024). The second is energy and social equity: the allocation of the costs of new infrastructure between large commercial users and households, and the consequences for energy poverty and energy justice (Carley & Konisky, 2020; Jenkins et al., 2016). The third is the sustainability of the digital transition itself: whether Europe can expand computing capacity in a manner compatible with its climate-neutrality commitments and with a just transition (Heffron & McCauley, 2018). These three dimensions form the common thread of the analysis. They link the sections on electricity demand and prices to those on energy poverty, environmental externalities, and the regulatory response of the European Union (EU).

The legal literature on data centres has grown quickly. Its centre of gravity is sustainability: the most complete doctrinal account of the EU data-centre acquis concludes that the current information regime should be converted into binding sustainability targets (Commins & Irion, 2025); the literature on sustainable AI regulation identifies environmental impact as a blind spot of AI governance built around trustworthiness and fundamental rights (Hacker, 2024; Perez Victorio et al., 2024); and recent work shows that the facility-level regime of the Energy Efficiency Directive and the model-level regime of the AI Act do not meet (Ebert et al., 2026). A parallel strand examines data centres as a source of system-stability risk and argues that EU law has completed the assessment stage of the risk-regulation cycle without beginning the management stage (Szuchy, 2026). The distributional consequences of data-centre expansion for households have received less attention. This article addresses that gap.

Against this background, the article pursues three aims. First, it characterises the scale and geography of data-centre electricity demand in Europe, with attention to the growing share attributable to AI. Second, drawing on three purposively selected European cases—Ireland, the United Kingdom, and Spain—it examines whether and how data-centre expansion has coincided with pressure on electricity systems and household costs, while distinguishing between contemporaneous trends and demonstrated causal effects. Third, it maps the EU's legal and policy response by legal status and asks how the sustainability costs and distributional effects of AI-related infrastructure should be incorporated into assessments of AI's social value. Section 2 presents the methodology, the rationale for case selection, and the terminology. Section 3 presents the evidence. Section 4 discusses the EU response and its energy-justice implications. Section 5 offers actor-specific recommendations, and Section 6 states the limitations of the study.

2. Methodology

2.1 Approach and Sources

The study adopted a qualitative approach that combined doctrinal legal analysis with a comparative, descriptive assessment of publicly available quantitative data. The doctrinal component examined the EU legal and policy instruments that bear on data-centre energy use and its social consequences. These include Directive (EU) 2023/1791 on energy efficiency (the recast Energy Efficiency Directive, EED), Delegated Regulation (EU) 2024/1364 establishing the EU data-centre rating scheme, the 2024 reform of the electricity market design, the Social Climate Fund, Regulation (EU) 2024/1689 (the AI Act), and the Commission's 2026 proposal for a Cloud and AI Development Act. The purpose of this component was to characterise the Union's response and to distinguish measures in force from measures formally proposed or merely under discussion.

The comparative component drew on data from international agencies (notably the IEA and the European Central Bank), EU bodies (the European Parliamentary Research Service, the European Network of Transmission System Operators for Electricity (ENTSO-E), and the Agency for the Cooperation of Energy Regulators (ACER)), national statistical offices and regulators (the Irish Central Statistics Office (CSO) and Commission for Regulation of Utilities (CRU), the United Kingdom's Department for Energy Security and Net Zero (DESNZ) and Department for Science, Innovation and Technology (DSIT), and the Spanish National Commission on Markets and Competition (CNMC)), and the peer-reviewed literature on data-centre energy use, energy poverty, energy justice, and electronic waste. Industry sources were used only where no official statistic exists—for installed data-centre

capacity in Spain and for the first results of the Spanish grid-capacity maps—and they are identified as such in the text. Consumer-organisation data were used for Spanish household bills, for which no official annual series is published.

2.2 Case Selection

Three national cases were selected purposively to capture distinct configurations of data-centre development in Europe. Ireland represents a highly concentrated market. It hosts a large share of European hyperscale capacity around Dublin, has the highest data-centre share of national electricity consumption in the EU, and has experienced explicit grid-capacity constraints and regulatory intervention in the connection process (Bresnihan & Brodie, 2021; CRU, 2025; CSO, 2026). The United Kingdom represents a mature but rapidly expanding market with a comparatively strong, interconnected grid. Although no longer an EU Member State, it shares the European market context and offers a comparator in which data-centre growth has so far coincided with stable or falling household bills. Spain represents an emerging data-centre location whose attraction rests on abundant renewable generation and available land, and in which rapid growth in connection requests is encountering grid-access saturation. Together, the three cases span the spectrum from established concentration to rapid emergence. They allow the same descriptive question—how data-centre expansion and household electricity costs have evolved together—to be examined under different structural conditions. The wider EU provides the common regulatory frame.

2.3 Period, Indicators, and Limitations

The comparison starts in 2023. The COVID-19 pandemic depressed demand and prices in 2020–2021, and the Russian invasion of Ukraine in February 2022 produced acute volatility in a European market that in 2021 still imported roughly 40%–45% of its natural gas from Russia. The year 2023 therefore offers the first post-crisis baseline. Two categories of indicator were traced for each case: (i) household electricity costs, measured as the median annual domestic bill for Ireland, the estimated typical annual domestic bill for the United Kingdom, and the average annual bill of a standard regulated-tariff consumer for Spain; and (ii) data-centre development, measured as metered electricity consumption (Ireland and the United Kingdom) and installed IT capacity (Spain). These indicators are not directly comparable across countries. Median and average bills, metered consumption, and installed capacity measure different things, and they are used descriptively, within each country, to characterise contemporaneous trends.

The study is analytical rather than experimental. It does not generate primary data and does not estimate causal effects. Household electricity prices in the period were shaped predominantly by wholesale gas prices, taxation, government subsidies, renewable generation, carbon prices, and regulated network charges. Where the text notes that data-centre growth coincided with price movements, the finding should be read as an association that warrants further investigation, not as a demonstrated causal contribution. The figures in Section 3 are descriptive illustrations of contemporaneous trends and are not statistical tests. Where a value is a projection or an industry expectation, it is marked as such in the figure and in the accompanying note.

2.4 Terminology: “Data Centres” and “AI Data Centres”

Much of the available statistical evidence refers to the data-centre sector as a whole rather than to electricity that can be attributed specifically to AI workloads. Following the IEA (2025), the term “data centres” is used when sector-wide statistics on consumption, capacity, or grid impact are reported, with the observation that AI workloads—run on accelerated servers—are an increasingly important and the fastest-growing component of that demand. The terms “AI data centres”, “AI workloads”, and “accelerated servers” are reserved for statements that the underlying source specifically supports, such as estimates of accelerated-server consumption or of the energy used in model training and inference (de Vries, 2023; Luccioni et al., 2024). The distinction matters for the argument: the sustainability and distributional concerns discussed below arise from data-centre expansion as such, while AI is the principal reason that this expansion is now accelerating. The phrase “AI data centres” in the title is used in this broader sense, to denote the data-centre expansion that AI is driving.

3. Results

3.1 Rising Electricity Demand of Data Centres: Global and European Trends

Data centres consumed about 415 TWh in 2024, roughly 1.5% of global electricity, and the IEA projects that this will more than double by 2030, to around 945 TWh or just under 3% of global electricity consumption (IEA, 2025). Growth accelerated further in 2025, when global data-centre electricity demand rose by approximately 17% (IEA, 2026). Earlier concern that data-centre consumption would grow in step with computing output had been

tempered by evidence that efficiency gains kept sector-wide demand roughly flat during the 2010s (Masanet et al., 2020; Mytton & Ashtine, 2022). AI has changed that picture. Accelerated servers equipped with specialised chips for AI workloads are proliferating. The IEA projects their electricity use to grow by about 30% per year in its base case, far outpacing conventional servers, and attributes almost half of the projected net increase in global data-centre electricity consumption to 2030 to such servers (de Vries, 2023; IEA, 2025). In the United States, data centres consumed an estimated 176 TWh in 2023 (4.4% of national electricity) and are projected to reach 325–580 TWh (6.7%–12%) by 2028, with AI-accelerated servers the principal driver (Shehabi et al., 2024).

Europe mirrors these trends and in places exceeds them. Data centres in the EU consume roughly 3% of the Union's electricity, about double the global share (Widuto, 2025). ENTSO-E identifies more than 10,500 facilities with at least 50 kW of IT power across Europe, with some 12.7 GW of installed IT capacity, of which 9.9 GW is located in the EU-27, and projects demand growth of more than 50% between 2025 and 2030 (ENTSO-E, 2026a). EU data-centre electricity demand could grow by around 70% by 2030, an additional roughly 45 TWh (IEA, 2025). Figure 1 shows these trajectories. The demand is highly concentrated: in Ireland, data centres accounted for 23% of metered electricity consumption in 2025 (CSO, 2026), and clusters around Dublin, Amsterdam, Frankfurt, and Madrid place pressure on local grids. A single large data centre can consume as much electricity in a year as around 100,000 households (Widuto, 2025), and independent measurements indicate that generating text or images with large models consumes orders of magnitude more energy per request than conventional classification tasks (Luccioni et al., 2024).

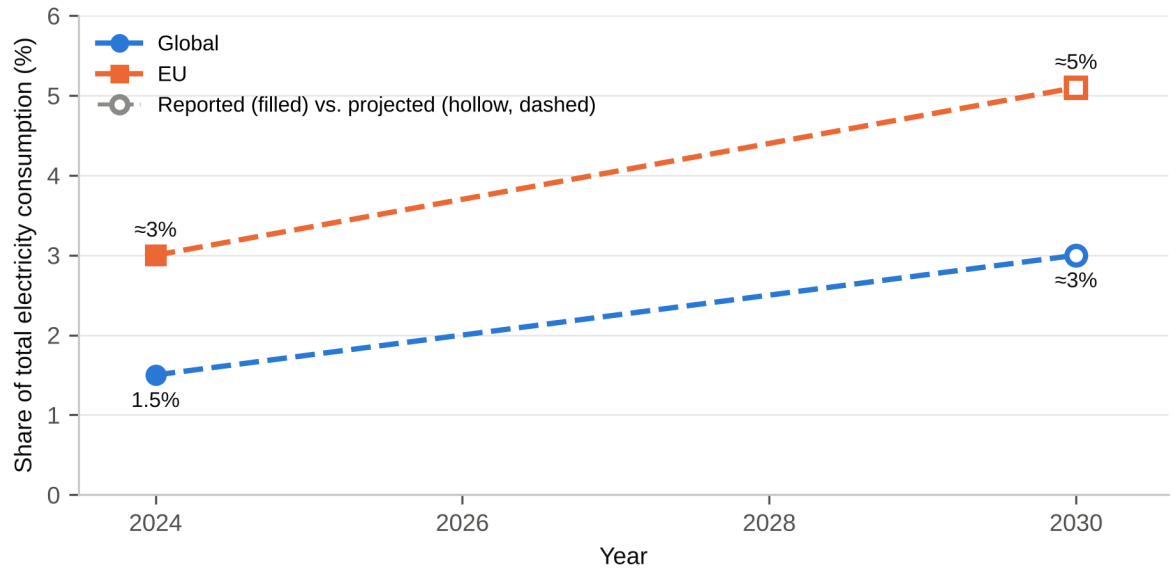


Figure 1. Data-centre share of total electricity consumption, global and EU, 2024 and 2030

Note: EU, European Union. Filled markers denote reported values for 2024; hollow markers and dashed lines denote published projections for 2030. The figure is a descriptive illustration of published trajectories, not a forecast produced by this study. Source: authors' illustration based on IEA (2025) and Widuto (2025); the 2030 EU value applies the IEA's projected 70% growth in EU data-centre demand to the current share, assuming that total consumption grows more slowly than data-centre demand.

The drivers of this growth are structural. Generative models, including large language models and image generators, require substantial computing power. Training a single leading model can consume energy on the order of a gigawatt-hour (Chen, 2025), and inference at scale—thousands of accelerators running continuously—accounts for a growing share of lifecycle energy use (Luccioni et al., 2024). Facilities also require energy for cooling and supporting infrastructure: the share of cooling in total consumption ranges from about 7% in efficient hyperscale facilities to over 30% in less efficient enterprise data centres (IEA, 2025). As AI adoption accelerates, so does the electricity drawn from the grid to sustain it.

3.2 Powering Data Centres: Renewable Energy Versus Fossil Fuels

A key question is how this expansion is being fuelled. Globally, data centres are supplied by a mix in which fossil fuels still dominate. In 2024, coal accounted for about 30% of the electricity supplied to data centres, renewables for about 27%, natural gas for about 26%, and nuclear for about 15% (IEA, 2025), as shown in Figure 2. Many technology companies have pledged to purchase renewable power or certificates to substantiate claims of "green" AI, and the EU has set the objective that data centres should be climate-neutral by 2030 (Widuto, 2025).

Interest in nuclear supply, including small modular reactors, is also growing (Widuto, 2025).

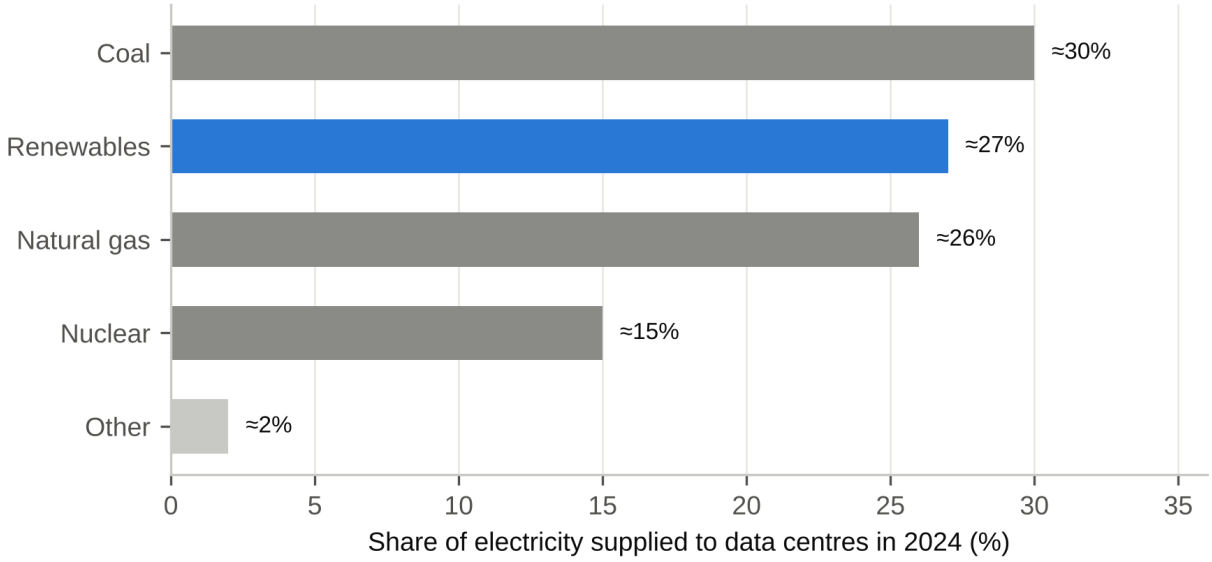


Figure 2. Global electricity supply mix for data centres in 2024, by source
Note: Values are approximate shares of electricity supplied to data centres, as reported in the IEA base case. Fossil fuels (coal and natural gas) together supply roughly 56%; renewables supply about 27% but are projected to meet nearly half of the growth in data-centre demand to 2030. Source: authors' chart based on IEA (2025).

Practice lags aspiration. When demand peaks or output from variable renewables falls, data centres draw on the prevailing grid mix, which in many regions still includes gas- or coal-fired generation. The IEA expects renewables to be the fastest-growing source for data centres, meeting nearly half of the additional demand to 2030, followed by natural gas and coal. On this trajectory, CO_2 emissions from electricity supplied to data centres peak at around 320 million tonnes by 2030 (IEA, 2025). Analysts at the European Central Bank likewise expect the surge in AI-related demand to be met by renewables and by natural gas "on the margin". They estimate that, in a hypothetical scenario in which all new AI-driven electricity needs to 2026 were met by gas, global gas prices could be some 7%–9% higher than otherwise (Burian & Stalla-Bourdillon, 2025). Supplying the expansion entirely from renewables would avoid this outcome but requires substantial investment in clean generation and grid capacity. The realistic near-term scenario is therefore mixed.

The EU has moved to improve transparency in data-centre energy use. Under Article 12 of the recast EED, owners and operators of data centres with an installed IT power demand of at least 500 kW must report key performance indicators—including energy consumption, renewable-energy use, and water consumption—annually to a European database (EU, 2023a). Delegated Regulation (EU) 2024/1364 established the first phase of a common Union rating scheme for data centres, with first reporting due in September 2024 (EU, 2024a). The first assessment of the reported data, published by the Commission in July 2025, captured some 770 facilities, by the Commission's own estimate only about one third of those obliged to report (Szuchy, 2026). The transparency regime is therefore in place but incomplete. Because renewables are projected to supply nearly half of data-centre demand growth (IEA, 2025), a meaningful share of new renewable capacity will be absorbed by the sector. Whether that supply is additional, or merely reallocated from other uses, is a central question for Europe's decarbonisation.

3.3 Impact on Electricity Grids and Prices

Beyond climate concerns, data-centre expansion places demands on power infrastructure that can translate into costs for other users. Network and system costs are recovered through regulated tariffs. Additional large loads that require new generation or grid reinforcement therefore raise costs for all users unless the incremental costs are allocated to those who cause them (ACER, 2025; Martin & Peskoe, 2025). The United States, where the AI build-out accelerated first, offers early evidence. In Virginia, the largest data-centre market, the state legislature's audit commission found that unconstrained demand would double within ten years, with data centres as the main driver; that current rates allocate costs appropriately; but that generation- and transmission-related costs for a typical residential customer could rise by an estimated USD 14–37 per month, in constant prices, by 2040 unless cost allocation changes (JLARC, 2024). Legal analysis of utility filings documents the mechanisms through which data-centre-driven investment is recovered from the general customer base (Martin & Peskoe, 2025).

The European experience is comparable in kind, if not yet in scale. In Ireland, the concentration of data centres around Dublin led the system operator to warn of capacity constraints, and data-centre metered electricity consumption rose by 20% in a single year (2022–2023) (CSO, 2024a). The regulator responded with a connection policy that conditions new data-centre connections on on-site or proximate generation and on renewable procurement (CRU, 2025). In the Netherlands, network congestion and the allocation of scarce connection capacity have become contested regulatory questions (van Schendel & Reins, 2024). Although European grids are generally stronger and more interconnected than those in the United States, the European Central Bank observes that heavy geographic clustering of data centres can create localised shortages and price spikes, and that limited cross-border capacity means that countries such as Ireland or the Netherlands cannot always import sufficient additional power when local demand surges (Burian & Stalla-Bourdillon, 2025). ENTSO-E has likewise described data centres as systemically relevant electricity users whose behaviour increasingly affects power-system operation (ENTSO-E, 2026a). Where data centres are deployed therefore matters as much as how many are deployed. In exposed regions, data-centre growth is associated with additional pressure on electricity systems and may contribute to upward pressure on network and system costs.

3.4 Country Evidence: The United Kingdom

During the energy crisis, the cost of energy to households in the United Kingdom was substantially masked by the government's Energy Price Guarantee (EPG), under which payments were made to suppliers between October 2022 and March 2025 to discount domestic prices (UK DESNZ, 2025a). For January–March 2023 the regulator's price cap for a typical dual-fuel household stood at £4,279 (Ofgem, 2022). The EPG limited the average bill to £2,500, with the difference paid from public funds (UK DESNZ, 2025a).

As wholesale prices fell, the estimated typical domestic electricity bill declined from approximately £1,213 in 2023 to £1,107 in 2024 and £1,063 in 2025, a modest downward and broadly stabilising trend (UK DESNZ, 2025b). Data-centre expansion did not follow this path. Official statistics compiled from meter-level data show that data centres in Great Britain consumed 4.1 TWh in 2023 and 4.5 TWh in 2024, about 2% of electricity drawn from the grid, an increase of 41% since 2020 (UK DESNZ, 2026). The government estimates colocation capacity at approximately 1.6 GW in 2024 (DSIT, 2025) and has projected, on a preliminary basis, 3.3–6.3 GW by 2030 depending on policy interventions (Clark, 2025), while the system operator expects data-centre consumption to rise from about 5 TWh to 22 TWh by 2030 (NESO, 2024). In September 2024 the government designated data centres as critical national infrastructure (UK Government, 2024), and in 2025 it invited applications for "AI Growth Zones" able to host at least 500 MW of AI infrastructure (Clark, 2025). The UK case therefore shows that rapid data-centre growth can coincide with falling household bills when wholesale prices fall. In this period, fuel costs and government policy, not data-centre load, dominated the movement of bills.

3.5 Country Evidence: Ireland

Ireland is the most concentrated data-centre market in the EU. Scholars have linked this position to the co-location of hyperscale investment with the country's wind-energy ambitions and to the political economy of a small state hosting global digital infrastructure (Bresnihan & Brodie, 2021). During the crisis period, the government subsidised household electricity bills through the Electricity Costs Emergency Benefit Scheme, with credits applied automatically to domestic meter accounts. These credits amounted to €550 in 2023 (three instalments), €400 in 2024 (three instalments), and €125 in 2025 (a single, final instalment) (Citizens Information, n.d.; CSO, 2024b; SEAI, n.d.).

Even with this support, household bills rose over the period. The median bill was approximately €880 in 2023 and about €1,000 in 2024 (CSO, 2024b; CSO, 2025), and the final credit of €125 in 2025 was far smaller than those of the two preceding years. Over the same years, data-centre electricity consumption rose markedly. Metered consumption increased by 20% between 2022 and 2023, to 6,334 gigawatt-hours (GWh); by a further 10% to 6,973 GWh in 2024; and by another 10% to 7,663 GWh in 2025, when data centres accounted for 23% of all metered electricity (CSO, 2024a; CSO, 2026). Figure 3 shows the two trends side by side. Their coincidence does not establish that data-centre demand raised household bills; the withdrawal of credits, wholesale gas prices, and network charges were the documented drivers of the increase. What the Irish case does show is that concentration produced binding grid constraints, and that these constraints have prompted regulatory conditioning of new connections. Under the CRU's December 2025 connection policy, applicants with a maximum import capacity above 10 megavolt-amperes (MVA) must provide dispatchable on-site or proximate generation or storage at least equal to their maximum import capacity and must match at least 80% of their annual demand with additional Irish renewable generation within six years; applicants between 1 and 10 MVA must provide on-site generation sized to their import capacity (CRU, 2025; Szuchy, 2026).

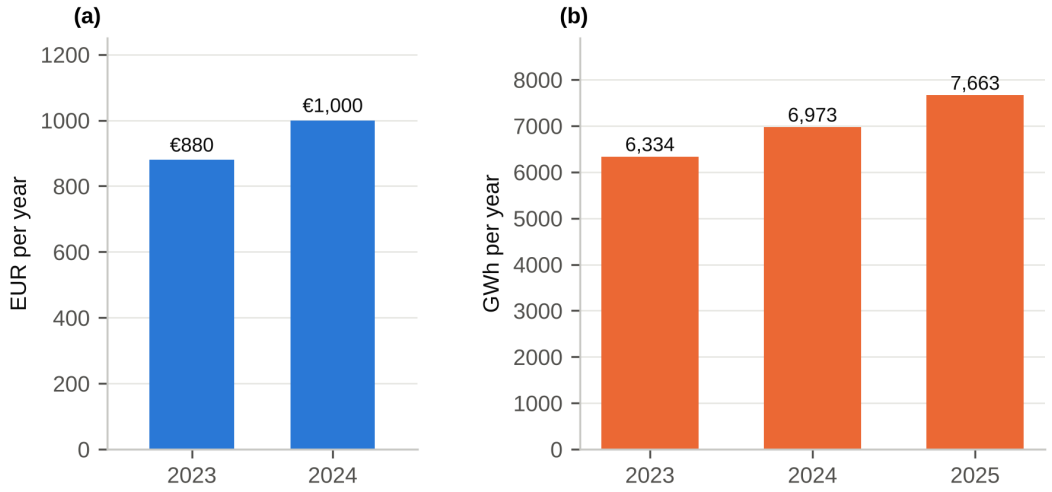


Figure 3. Ireland, 2023–2025: (a) median annual household electricity bill after government credits; (b) data-centre metered electricity consumption

Note: The two panels show different measures on separate scales and illustrate contemporaneous trends only; they are not a test of any statistical or causal relationship. Panel (a) shows official median bills for 2023 and 2024; no official figure for 2025 had been published at the time of writing. Source: authors' chart based on CSO (2024a); CSO (2024b); CSO (2025); CSO (2026).

3.6 Country Evidence: Spain

In the EU's day-ahead electricity market, the wholesale price is set by the marginal plant needed to meet demand. In Spain, gas-fired plants frequently set this price despite high renewable output, so that the benefit of low-cost renewable generation is only partly passed through to consumers; the 2022 crisis made this pass-through problem a central issue of European market reform (Fabra, 2023). A significant portion of the household bill—including the bill of households on the regulated tariff for small consumers (Precio Voluntario para el Pequeño Consumidor, PVPC)—consists of regulated costs, including network tolls and system charges. In addition, fossil-fuel plants must purchase CO_2 allowances under the EU Emissions Trading System, so that rising carbon prices feed through to consumer prices. Temporary tax reductions introduced during the crisis, including the reduced value-added tax rate on electricity, were maintained through 2024 and expired at the start of 2025 (Jefatura del Estado, 2023).

Following the extreme price spikes of 2022, Spanish household bills were stabilised by temporary tax and levy reductions before rising again in 2025. According to the consumer organisation FACUA, the annual bill of a standard PVPC household (4.4 kW contracted power, 366 kWh per month) rose by 15.5% in 2025, from €845 to €976 (an average of €81.32 per month), making 2025 the third most expensive year on record after 2022 and 2021 (FACUA, 2026). Two documented developments contributed to this increase. The first was the expiry of the temporary tax reductions at the beginning of 2025, which returned the value-added tax rate on electricity to its standard level (Jefatura del Estado, 2023). The second was the Iberian blackout of 28 April 2025 (ENTSO-E, 2026b), after which the system operator applied a "reinforced" scheduling of the system to mitigate rapid voltage variations. By the operator's own account, this reinforced operation accounted for 2.34% of total system costs, about €422 million, between May and October 2025, a cost that is passed through to consumers in the final price of energy (Red Eléctrica, 2025).

Over the same period, Spain has become Southern Europe's leading hub for data centres, attracting investment on the strength of abundant and comparatively cheap renewable electricity, connectivity, available land, and supportive national policy, including the first EU AI supervisory agency and an updated national AI strategy (OECD, 2025). Installed IT capacity in the Madrid region, which hosts about 55% of national capacity, rose from 147 MW in 2023 to 194.5 MW in 2024, an increase of 32%, and the industry association's business-as-usual scenario projects national capacity of about 2,180 MW beyond 2027 (Spain DC, 2024), as shown in Figure 4. Grid access has become the binding constraint. Under CNMC Circular 1/2024, distribution system operators were required to publish capacity maps for new demand connections; when the maps were first published in September 2025, the first results compiled by the distribution companies' association indicated that about 83% of distribution-network nodes had no available capacity for new demand (Aelec, 2025; CNMC, 2024). As in Ireland, the 2025 bill increase is accounted for by documented tax, regulatory, and system-security factors. The relevance of the Spanish case lies instead in the speed with which new load is absorbing renewable additions and saturating grid access, which raises the prospect of future network-reinforcement costs whose allocation will matter for households.

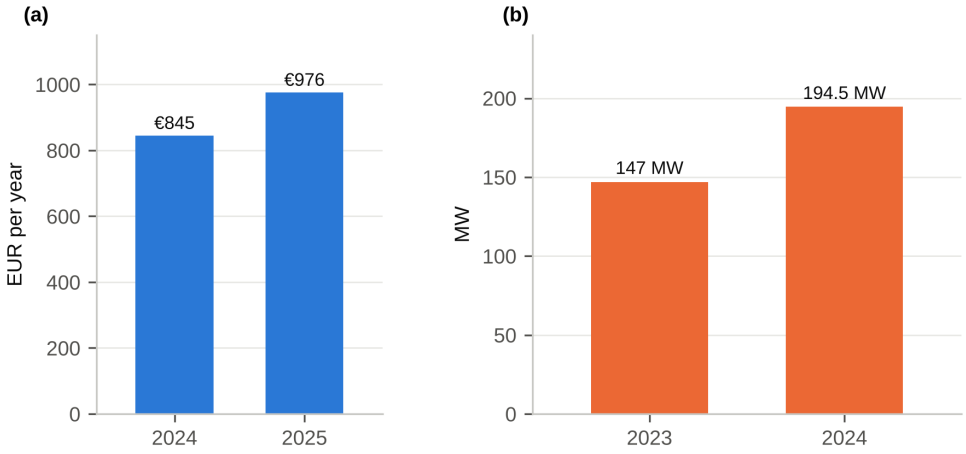


Figure 4. Spain, 2023–2025: (a) average annual electricity bill of a standard regulated-tariff household; (b) installed data-centre IT capacity in the Madrid region

Note: The two panels show different measures on separate scales and illustrate contemporaneous trends only; they are not a test of any statistical or causal relationship. Panel (a) refers to the consumer organisation FACUA's standard user on the regulated average household electricity bill tariff (4.4 kW contracted power, 366 kWh per month); a comparable annual figure for 2023 is not published by the same source. Panel (b) refers to the Madrid region, which hosts about 55% of Spanish capacity. Source: authors' chart based on FACUA (2026) and Spain DC (2024).

Table 1. Comparative synthesis of the three country cases (2023–2025)

Dimension	Ireland	United Kingdom	Spain
Data-centre trend	Metered consumption 6,334 GWh (2023) → 6,973 GWh (2024) → 7,663 GWh (2025); 23% of national metered electricity in 2025.	Metered consumption 4.1 TWh (2023) → 4.5 TWh (2024), about 2% of grid electricity; colocation capacity ≈1.6 GW (2024), projected 3.3–6.3 GW by 2030.	Madrid-region installed IT capacity 147 MW (2023) → 194.5 MW (2024); ≈83% of distribution-network nodes without capacity for new demand (Sept. 2025).
Household electricity-cost context	Median bill ≈€880 (2023) → ≈€1,000 (2024) as government credits (€550 → €400 → €125) were withdrawn.	Typical bill £1,213 (2023) → £1,107 (2024) → £1,063 (2025); falling and stabilising after the withdrawal of the Energy Price Guarantee.	Standard regulated-tariff bill €845 (2024) → €976 (2025), +15.5%; third most expensive year on record after 2022 and 2021.
Documented drivers of bill movements	Withdrawal of credits; wholesale gas prices; network charges.	Wholesale gas prices; price-cap methodology; withdrawal of EPG support.	Expiry of temporary tax reductions; regulated tolls and charges; cost of reinforced system operation after the April 2025 blackout; carbon prices.
Main sustainability concern	Grid capacity and security of supply; concentration around Dublin; reliance on gas-fired back-up.	Pace of expansion relative to grid reinforcement; land and power constraints around London.	Absorption of renewable additions by new load; grid-access saturation; water and land use.
Policy relevance	CRU connection policy (Dec. 2025): on-site or proximate generation and 80% Irish renewables for applicants above 10 MVA (in force).	Outside the EU framework; data centres designated critical national infrastructure (2024); AI Growth Zones; connection-queue reform.	EU framework applies (EED reporting; rating scheme); CNMC Circular 1/2024 capacity maps; national AI strategy.
What the case illustrates	Concentration → binding grid constraints → regulatory conditioning of new connections.	Rapid expansion can coincide with falling bills when fuel costs fall; bills dominated by wholesale prices and policy.	Emerging hub in which renewable abundance attracts load; grid-access saturation and regulated charges shape bills.

Note: CRU, Commission for Regulation of Utilities; EU, European Union; EED, Energy Efficiency Directive; CNMC, National Commission on Markets and Competition. Median and average bills, metered consumption, and installed capacity are different measures and are not comparable across columns; figures are as reported by the sources cited in Sections 3.4–3.6.

3.7 Comparative Synthesis

Table 1 draws the three cases together. Read jointly, they support four observations. First, data-centre

development expanded rapidly in all three countries throughout 2023–2025, irrespective of the direction of household bills. Second, the movement of household bills in the period is explained in each case by documented factors other than data-centre load: the withdrawal of crisis-era support in Ireland and the United Kingdom, the reversion of taxes and the post-blackout system costs in Spain, and wholesale gas prices everywhere. The cases therefore do not, on their own, demonstrate a price effect of data-centre expansion. Third, the channel through which data-centre growth most clearly affects electricity systems is capacity rather than price: binding grid constraints in Ireland, saturated distribution nodes in Spain, and the pace of reinforcement in the United Kingdom. Fourth, this capacity channel carries the distributional risk, because reinforcement and system-security costs are recovered through regulated tariffs from all users unless they are allocated otherwise. The Irish regulator's 2025 connection policy shows that this allocation question is already being answered in practice by conditioning connections on self-supply and renewable procurement. The wider European question is whether such conditions become the norm.

3.8 The Burden on Vulnerable Households: Energy Poverty and the Risk of a "Double Burden"

Higher electricity prices fall hardest on low-income households. Energy poverty—the inability to afford adequate energy services, such as keeping a home adequately warm—is a persistent and, since the energy crisis, growing problem in the EU. In 2023, 10.6% of the EU population were unable to keep their homes adequately warm, up from 6.9% in 2021 (Eurostat, 2024). The 2022 price shock increased household energy costs unevenly: the largest proportional burdens fell on lower-income households, which spend a greater share of their income on energy and have fewer options to adapt (Guan et al., 2023). Research on European energy poverty stresses that vulnerability is structural—linked to housing quality, tenure, income, and tariff design—so that any sustained upward pressure on system costs is transmitted disproportionately to those already at risk (Bouzarovski et al., 2021). The recast EED now supplies the Union's first binding definition of energy poverty, and the 2024 market-design reform strengthened protection against disconnection for vulnerable customers (EU, 2019c; EU, 2023a).

It is important to distinguish what the evidence in this article shows from what it implies. The country cases show observed energy and infrastructure pressures: rapidly growing data-centre load, binding grid constraints in the most concentrated market, and saturated grid access in the fastest-growing one. What follows from them is a distributional inference. If reinforcement and system-security costs arising from these loads are recovered through general tariffs, households will bear part of them, and households in energy poverty will bear them least easily. The cases do not measure such a transfer, and the "double burden" is presented here as a risk rather than as a demonstrated effect. The risk has two sides. On the benefit side, a digital divide persists: wealthier individuals, firms, and countries can more readily access and benefit from AI, whereas poorer communities may lack the tools, skills, or infrastructure to do so (Lythreatis et al., 2022). On the cost side, low-income households devote a larger share of income to energy, so any increment in system costs passed through tariffs weighs more heavily on them (Guan et al., 2023). The combination—least benefit and greatest relative exposure to cost—is what makes the fairness question acute.

The energy-justice literature offers the conceptual vocabulary for this question. Distributional justice concerns the fair allocation of the benefits and burdens of energy systems; procedural justice concerns who participates in the decisions that allocate them; and recognition justice concerns whether the needs of vulnerable groups are acknowledged at all (Jenkins et al., 2016; Sovacool & Dworkin, 2015). Research on the clean-energy transition warns that transitions can reproduce or deepen existing inequities unless their distributional consequences are managed explicitly (Carley & Konisky, 2020). Economic research on automation indicates that its gains and losses are distributed unevenly across workers and countries (Acemoglu & Restrepo, 2022; Cazzaniga et al., 2024). Regulators in Europe and the United States are accordingly considering connection conditions, dedicated tariff classes, and contributions designed to ensure that large data users pay a fair share of the costs they cause (ACER, 2025; CRU, 2025; JLARC, 2024). These measures are examined in Section 4.

3.9 Environmental Costs: Carbon Emissions and Electronic Waste

Beyond electricity bills, data-centre expansion carries environmental externalities, chiefly carbon emissions and electronic waste. Even where partly supplied by renewables, data centres retain a substantial carbon footprint because of the fossil share of the grid mix and the carbon embodied in hardware. An early peer-reviewed estimate warned that by 2027 AI-related electricity consumption alone could reach 85–134 TWh per year, comparable to the annual consumption of the Netherlands (de Vries, 2023), and the IEA projects that CO_2 emissions from electricity supplied to data centres will peak at around 320 million tonnes by 2030 (IEA, 2025). The EU's objective of climate-neutral data centres by 2030 is intended to address this. Aligning the sector's trajectory with climate goals nevertheless remains time-critical, and the lack of transparency about the environmental costs of generative AI has itself been identified as a barrier to governance (Crawford, 2024; Kaack et al., 2022).

Less discussed, but increasingly urgent, is the question of hardware lifecycles and electronic waste. AI systems

rely on specialised accelerators and servers with shorter replacement cycles than conventional IT equipment, as operators refresh hardware to maintain performance. A recent peer-reviewed assessment estimates that generative AI could generate a cumulative 1.2–5.0 million tonnes of e-waste over 2020–2030, and that circular-economy strategies along the value chain—extended server lifetimes, reuse, remanufacturing, and recycling—could reduce this by 16%–86% (Wang et al., 2024). For perspective, the upper estimate is equivalent to roughly 8% of the 62 million tonnes of e-waste generated worldwide in the single year 2022. Global e-waste is projected to reach 82 million tonnes by 2030, and only 22.3% of it was documented as formally collected and recycled in 2022 (Baldé et al., 2024). The AI-related stream is therefore modest in aggregate but fast-growing, hazardous, and rich in critical raw materials, and it is largely absent from current waste-management planning.

The impact extends upstream. Manufacturing high-performance chips and servers is energy- and resource-intensive. Lifecycle analyses show that embodied (manufacturing) emissions account for a substantial share of computing's carbon footprint and, for hardware operated on low-carbon grids, a dominant share (Gupta et al., 2022). A fast upgrade cycle therefore implies more mineral extraction, more manufacturing emissions, and more discarded electronics. European policymakers are beginning to acknowledge this dimension. The Commission's 2025 evaluation of the Waste Electrical and Electronic Equipment (WEEE) Directive found that the Directive does not adequately address new waste streams from digital technologies that are rich in critical raw materials, and that only about 40% of WEEE is recycled in the EU (European Commission, 2025d). On current trajectories, AI hardware could become a significant contributor to hazardous-waste volumes by the end of the decade unless circular-economy obligations are extended to it.

4. Discussion

4.1 The European Union Response, by Legal Status

The EU is developing a multi-instrument response that seeks to balance AI innovation with sustainability and social responsibility. The instruments involved do not, however, share the same legal or institutional status. It is therefore important to distinguish binding measures in force from measures formally proposed and from options under discussion. Table 2 classifies them accordingly.

Three features of this landscape stand out. First, the binding measures in force are predominantly measures of transparency and consumer protection. The EED reporting obligation and the rating scheme make data-centre energy performance visible; the AI Act makes model-level energy consumption documentable; and the market-design reform strengthens the protection of vulnerable customers. None of these measures allocates the costs of new load, and the Social Climate Fund, although substantial, is designed for the social incidence of carbon pricing in buildings and road transport rather than for data-centre-driven demand growth (Szuchy, 2026). Second, the measure most directly aimed at expansion—the proposed Cloud and AI Development Act (European Commission, 2026), which gives legislative form to the capacity ambition of the AI Continent Action Plan (European Commission, 2025b)—is a growth instrument with sustainability conditions attached. Its final form will determine whether acceleration zones and shortened permitting are matched by binding requirements on clean supply and grid-connection cost allocation. Third, the question of who pays for reinforcement is being answered first at national level, in Ireland's connection policy, rather than through EU law. Article 18 of Regulation (EU) 2019/943 provides the legal basis for cost-reflective network charges (EU, 2019c), and ACER has articulated principles of cost-reflectivity and non-discrimination, warning that unjustified exemptions, discounts, and à la carte tariff regimes for particular user groups shift costs to other users (ACER, 2025). Tariff design nevertheless remains a national competence, and the Commission's December 2025 guidance on grid connections concedes that no explicit prioritisation framework based on economic criteria yet exists (European Commission, 2025c).

4.2 Cost-Causer Pays as a Question of Energy Justice

The "cost-causer pays" principle—that the loads which necessitate new generation and network investment should bear its incremental cost—is usually discussed as a matter of tariff design. Read through the energy-justice framework introduced in Section 3.8, it is also a matter of distributional justice. Network and system costs are among the components of household bills least amenable to consumer choice, and they are regressive in incidence because they are largely unrelated to income (Bouzarovski et al., 2021; Guan et al., 2023). If the costs of accommodating hyperscale load are socialised through these components, the transfer runs from households, including energy-poor households, to large commercial users. If the costs are allocated to the connecting load, it does not. The choice between these outcomes is therefore a choice about whether the expansion of digital infrastructure is distributed fairly between large commercial users and households (Carley & Konisky, 2020; Jenkins et al., 2016). The just-transition literature adds a procedural dimension: communities hosting infrastructure, and consumer representatives, should have a voice in the connection and tariff decisions that allocate these costs (Heffron & McCauley, 2018). Ireland's 2025 connection policy and the Virginia findings show that regulators can

operationalise the principle through connection conditions and tariff classes (CRU, 2025; JLARC, 2024). The EU's contribution could be to make cost-reflective allocation for very large loads an explicit expectation under the electricity market framework, building on ACER's tariff principles (ACER, 2025), so that the benefits of AI-driven investment are not purchased with a hidden charge on those least able to pay.

Table 2. EU measures relevant to data-centre energy use and its social consequences, by legal status

Instrument	Status	Relevance to the Present Analysis
Directive (EU) 2023/1791 on energy efficiency (recast EED) (EU, 2023a)	In force (October 2023)	Art. 12: annual reporting of data-centre energy performance (≥500 kW IT demand) to a Union database; Art. 8(3): a share of the energy-savings obligation must benefit people affected by energy poverty and vulnerable customers; Art. 2(52): first binding Union definition of energy poverty.
Delegated Regulation (EU) 2024/1364 (EU data-centre rating scheme) (EU, 2024a)	In force (first reporting September 2024)	Common indicators—energy use, renewable share, water use, waste-heat reuse—enabling comparison of data-centre sustainability performance; first assessment (July 2025) captured about one third of obliged facilities.
Regulation (EU) 2023/955 (Social Climate Fund) (EU, 2023b)	In force; operational from 2026	Funding for Member States' measures supporting vulnerable households in the context of carbon pricing for buildings and road transport (ETS2); data-centre-driven demand growth is not within its remit.
Regulation (EU) 2024/1747 and Directive (EU) 2024/1711 (electricity market design reform) (EU, 2024b; EU, 2024d); Regulation (EU) 2019/943, Art. 18 (EU, 2019c); Directive (EU) 2019/944, Arts. 6a and 28a (EU, 2019b)	In force	Cost-reflective, transparent, and non-discriminatory network charges (the legal basis for allocating reinforcement costs to the loads that cause them); flexible, non-firm connection agreements in congested areas; strengthened protection of vulnerable customers against disconnection.
Regulation (EU) 2024/1689 (AI Act) (EU, 2024c)	In force; staged application 2025–2027	Annex XI requires providers of general-purpose AI models to document known or estimated energy consumption; Art. 40 allows standardisation requests on the energy-efficient development of AI systems.
Directive 2012/19/EU (WEEE) (EU, 2012); Regulation (EU) 2019/424 (ecodesign for servers and data storage) (EU, 2019a); Regulation (EU) 2024/1781 (Ecodesign for Sustainable Products) (EU, 2024e)	In force	Existing framework for the collection, treatment, and design of electronic equipment; data-centre hardware is covered in general terms but without AI-specific provisions.
Cloud and AI Development Act (European Commission, 2026)	Formally proposed (COM(2026) 502, 3 June 2026); pending before Parliament and Council	Data-centre acceleration zones with a maximum 12-month permitting timeline, sustainability conditions, preference for brownfield sites, and conditions for grid connection; gives legislative form to the ambition to at least triple EU data-centre capacity within five to seven years.
AI Continent Action Plan (European Commission, 2025b); Action Plan for Affordable Energy (European Commission, 2025a); Commission guidance on efficient and timely grid connections (European Commission, 2025c)	Announced policy and guidance (2025); non-binding	Capacity ambition; announced data-centre energy-efficiency package; guidance recommending "first-ready, first-served" connection queues and grid acceleration areas, while noting the absence of any explicit prioritisation framework.
Revision of the WEEE Directive (European Commission, 2025d)	Announced; evaluation published July 2025; revision prepared within the forthcoming Circular Economy Act	Evaluation found inadequate coverage of new digital, critical-raw-material-rich waste streams and fragmented extended producer responsibility.
Cost-allocation ("cost-causer pays") reforms for large loads; connection conditioning	Under discussion at EU level (ACER, 2025); adopted nationally in Ireland (CRU, 2025)	Allocation of reinforcement and system-security costs to large loads rather than to residential users; on-site generation and renewable-procurement conditions for connections.
Extended producer responsibility for data-centre hardware; AI sustainability labels or metrics beyond the AI Act	Under discussion; no formal proposal	Options for internalising end-of-life and lifecycle costs of AI hardware and for making the energy and carbon performance of models and facilities comparable.

Note: EU, European Union; EED, Energy Efficiency Directive; WEEE, Waste Electrical and Electronic Equipment. Classification reflects the position at the time of writing (November 2026). "In force" denotes adopted Union law, some provisions of which may apply in stages; "formally proposed" denotes a Commission legislative proposal pending in the ordinary legislative procedure; "announced" denotes a non-binding policy document or guidance; "under discussion" denotes options debated by institutions, regulators, or the literature without a formal EU proposal.

4.3 Incorporating Sustainability Costs and Distributional Effects Into Assessments of AI's Social Value

This study does not measure the overall social value of AI, and it does not propose a framework capable of calculating it. Its evidence bears on a narrower but tractable question: how the sustainability costs and distributional effects of AI-related infrastructure should enter such assessments. At present they largely do not. Assessments of AI's economic contribution are typically framed in terms of productivity and output. The energy, carbon, hardware, and network costs documented above are external to them, and their distributional incidence is invisible. Three steps would bring these costs inside the assessment.

The first step is measurement. The EED reporting obligation and the AI Act's documentation requirement provide legal hooks for facility-level and model-level energy data (EU, 2023a; EU, 2024c), although the two regimes do not yet connect the facility to the workload it hosts (Ebert et al., 2026), and the IEA's scenario work provides a consistent method for converting consumption into emissions (IEA, 2025). Extending disclosure to embodied carbon and hardware end-of-life, in line with the lifecycle evidence (Gupta et al., 2022; Wang et al., 2024), would make the full environmental cost of a model or facility reportable rather than estimated. The second step is cost allocation. Regulatory decisions on connection conditions and network tariffs already determine who pays for reinforcement. Requiring a distributional impact assessment when very large loads connect—identifying the incidence of reinforcement and system-security costs across customer classes, including energy-poor households—would make the distributional effect explicit before it is incurred. The third step is the treatment of offsets. AI is increasingly used to improve energy efficiency in buildings and industry, to integrate renewables, and to optimise networks (IEA, 2025; Widuto, 2025). Such gains are a legitimate credit against AI's energy cost, but they should be evidenced and attributed at the level of specific applications rather than assumed for the sector as a whole. Where these three steps are taken, the question of whether a given expansion of AI infrastructure improves social welfare becomes answerable with evidence, and the EU's stated ambition that AI should "work for people" acquires a sustainability and equity test that can be applied.

5. Conclusions and Recommendations

AI is driving a rapid expansion of data-centre electricity demand that is geographically concentrated, still supplied predominantly from fossil sources at the global level, and accompanied by a growing stream of hardware waste. The three European cases show that the clearest channel through which this expansion affects electricity systems is capacity—binding grid constraints in Ireland, saturated grid access in Spain, and the pace of reinforcement in the United Kingdom—rather than a demonstrated effect on household prices, which in the period examined were dominated by gas prices, taxation, subsidies, and regulated charges. Because reinforcement and system-security costs are recovered through tariffs from all users unless allocated otherwise, the expansion nevertheless carries a distributional risk for households already in energy poverty. Five recommendations follow from this evidence, each addressed to the actors able to act on it.

Cost-reflective allocation of network and system costs (national energy regulators and grid operators): The evidence of geographically concentrated grid pressure in Ireland and Spain leads directly to the allocation question. Regulators should apply the cost-reflectivity requirement of Regulation (EU) 2019/943 to very large loads through connection conditions and dedicated tariff classes, following the Irish connection policy and the tariff principles articulated by ACER, so that reinforcement and system-security costs are not recovered from residential customers.

Additionality of clean supply (data-centre operators and energy regulators): Because renewables are projected to meet nearly half of data-centre demand growth while the current supply mix remains fossil-dominated, operators should be required—through connection conditions, the proposed Cloud and AI Development Act, or the rating scheme—to demonstrate that their renewable supply is additional, in the form of new capacity or long-term contracts for new build, rather than reallocated from other users, and to provide flexibility to the system.

Safeguards for vulnerable consumers (Member States and social-policy authorities): The energy-poverty evidence indicates that any measure likely to raise system costs should be accompanied by targeted support. Member States should direct the energy-savings obligation of the Energy Efficiency Directive towards vulnerable households in regions hosting large loads, should consider extending the logic of the Social Climate Fund to the incidence of infrastructure-driven cost increases, and should include a distributional impact assessment in connection and tariff decisions for very large customers.

Hardware circularity (operators, equipment manufacturers, and the EU legislator): The rapid replacement cycle of AI hardware and the projected growth of generative-AI electronic waste lead to the circular-economy

recommendation. The revision of the WEEE Directive within the Circular Economy Act should extend producer responsibility and design-for-reuse requirements explicitly to data-centre equipment, since the evidence indicates that circular strategies could substantially reduce this waste stream.

Sustainability metrics for models and facilities (the European Commission, standardisation bodies, and operators): Building on the EED database, the rating scheme, and the AI Act's documentation requirement, comparable disclosure of energy, carbon (operational and embodied), water, and end-of-life performance for large models and facilities would make the trade-offs visible and allow the offsets that AI provides to energy systems to be evidenced rather than assumed.

6. Limitations and Future Research

The study is descriptive and doctrinal. It does not estimate the causal effect of data-centre expansion on household prices, it relies for Spanish capacity on industry data, and it examines three cases whose indicators are not directly comparable. Its findings should be read as a structured account of contemporaneous trends and of the legal framework, and as a set of testable propositions. Future research should estimate the incidence of data-centre-driven network costs econometrically at regional level, evaluate the distributional effects of connection policies such as Ireland's, and quantify the embodied carbon and electronic waste of AI hardware deployed in Europe. With such evidence, and with the instruments now available to the Union, it is possible to expand computing capacity without exhausting grids, deepening energy poverty, or degrading the environment.

Author Contributions

Conceptualisation, R.S.; methodology, R.S.; investigation, R.S. and A.E.G.; data curation, A.E.G.; formal analysis, R.S. and A.E.G.; visualisation, A.E.G.; writing—original draft preparation, R.S. and A.E.G.; writing—review and editing, R.S.; supervision, R.S. Both authors have read and agreed to the published version of the manuscript.

Data Availability

The data supporting the findings of this study are derived from the publicly available sources cited in the references; the values plotted in Figures 1–4 are reported in Sections 3.1–3.6 and Table 1. No new datasets were generated during the study.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-Assisted Technologies

During the preparation and revision of this manuscript, the authors used a generative AI large language model (Anthropic, Claude) in 2026 solely for minor language editing and formatting assistance. The tool was not used to generate data, results, references, analysis, or substantive intellectual content, and it is not listed as an author. After using this tool, the authors reviewed and edited the content as required and take full responsibility for the accuracy, originality, integrity, and content of the published article.

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